

Chapter 8

Advanced Counting Techniques

8.1 - 8.2

Chapter Summary

- Applications of Recurrence Relations
- Solving Linear Recurrence Relations
 - *Homogeneous Recurrence Relations*
 - *Nonhomogeneous Recurrence Relations*

8.1. Recurrence Relations (RR)

Recurrence Relations

In Chapter 5 we discussed how sequences can be defined recursively.

Recall that a recursive definition of a sequence specifies:

1. **One** or **more initial terms** and
2. A **rule** for determining subsequent terms from those that proceed them. A **rule** of the latter sort (whether or not it is part of a recursive definition) is called a **recurrence relation**. such relations can be used in studying and **solving counting problems**.

Recurrence Relations

- A **recurrence relation** (R.R., or just **recurrence**) for the sequence $\{a_n\}$ is **an equation** that expresses a_n in terms of **one or more previous terms** of the sequence $a_0, \dots, a_{n-1}$, for all integers n with $n \geq n_0$, where n_0 is a nonnegative integer
 - I.e., just a recursive definition, without the base cases.
- A **sequence** is called a **solution** of a **recurrence relation** if its **terms satisfy the recurrence relation**.

Recurrence Relation

Example

- Consider the recurrence relation

$$a_n = 2a_{n-1} - a_{n-2} \quad (n \geq 2).$$

- Which of the following are solutions?

$$a_n = 3n$$

$$a_n = 2^n$$

$$a_n = 5$$

Recurrence Relation

Example

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$$a_n = 2a_{n-1} - a_{n-2} \quad (n \geq 2).$$

- Which of the following are solutions?

$$a_n = 3n \quad \text{Yes}$$

$$a_n = 2^n \quad \text{No}$$

$$a_n = 5 \quad \text{Yes}$$

Financial Application

Example: Suppose that a person deposits \$10,000.00 in a savings account at a bank yielding 11% per year with interest compounded annually. How much will be in the account after 30 years?

Let P_n denote the amount in the account after n years. P_n satisfies the following recurrence relation:

$$P_n = P_{n-1} + 0.11P_{n-1} = (1.11) P_{n-1}$$

with the initial condition $P_0 = 10,000$

Continued on next slide →

Financial Application

$$P_n = P_{n-1} + 0.11P_{n-1} = (1.11) P_{n-1}$$

with the initial condition $P_0 = 10,000$

Solution: Forward Substitution

$$P_1 = (1.11)P_0$$

$$P_2 = (1.11)P_1 = (1.11)^2 P_0$$

$$P_3 = (1.11)P_2 = (1.11)^3 P_0$$

⋮

$$P_n = (1.11)P_{n-1} = (1.11)^n P_0 = (1.11)^n 10,000$$

$$P_n = (1.11)^n 10,000$$

$$P_{30} = (1.11)^{30} 10,000 = \$228,992.97$$

8.2. SOLVING LINEAR RECURRENCE RELATIONS

K-LiHoReCoCo

General Solution Schemas

- A linear homogeneous recurrence of degree k with constant coefficients (“k-LiHoReCoCo”) is a rec. rel. of the form

$$a_n = c_1 a_{n-1} + \dots + c_k a_{n-k},$$

where the c_i are all real numbers, and $c_k \neq 0$.

- it is *linear* because the right-hand side is a sum of the previous terms of the sequence each multiplied by a function of n .
- it is *homogeneous* because no terms occur that are not multiples of the a_j s. Each *coefficient* is a *constant*.
- the *degree is k* because a_n is expressed in terms of the previous k terms of the sequence.

Examples of Linear Homogeneous Recurrence Relations **linear?** **Homogeneous? Which degree?**

- $P_n = (1.11)P_{n-1}$
- $f_n = f_{n-1} + f_{n-2}$
- $a_n = a_{n-1} + a_{n-2}^2$
- $H_n = 2H_{n-1} + 1$
- $B_n = nB_{n-1}$

Examples of Linear Homogeneous Recurrence Relations

- $P_n = (1.11)P_{n-1}$ linear homogeneous recurrence relation of degree one
- $f_n = f_{n-1} + f_{n-2}$ linear homogeneous recurrence relation of degree two
- $a_n = a_{n-1} + a_{n-2}^2$ not linear
- $H_n = 2H_{n-1} + 1$ not homogeneous
- $B_n = nB_{n-1}$ coefficients are not constants

Solving Linear Homogeneous Recurrence Relations

- The basic approach is to look for solutions of the form $a_n = r^n$, where r is a constant.
- Note that $a_n = r^n$ is a solution to the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$ if and only if $r^n = c_1 r^{n-1} + c_2 r^{n-2} + \dots + c_k r^{n-k}$.
- Algebraic manipulation yields the *characteristic equation*:
$$r^k - c_1 r^{k-1} - c_2 r^{k-2} - \dots - c_{k-1} r - c_k = 0$$
- The sequence $\{a_n\}$ with $a_n = r^n$ is a *solution* if and only if r is a *solution* to the *characteristic equation (C.E)*.
- The solutions to the characteristic equation are called the *characteristic roots of the recurrence relation*. The roots are used to give an explicit formula for all the solutions of the recurrence relation.

Solving Linear Homogeneous Recurrence Relations of Degree Two

Theorem 1: Let c_1 and c_2 be real numbers.

Suppose that $r^2 - c_1r - c_2 = 0$ has **two distinct roots** r_1 and r_2 ($r_1 \neq r_2$)

- Then the sequence $\{a_n\}$ is a solution to the recurrence relation $a_n = c_1a_{n-1} + c_2a_{n-2}$ if and only if
$$a_n = \alpha r_1^n + \alpha_2 r_2^n$$

for $n = 0, 1, 2, \dots$, where α_1 and α_2 are constants.

Using Theorem 1

Example: What is the solution to the recurrence relation

$$a_n = a_{n-1} + 2a_{n-2} \text{ with } a_0 = 2 \text{ and } a_1 = 7?$$

Solution: The characteristic equation is $r^2 - r - 2 = 0$.

Its roots are $r = 2$ and $r = -1$. Therefore, $\{a_n\}$ is a solution to the recurrence relation if and

only if $a_n = a_1 2^n + a_2 (-1)^n$, for some constants a_1 and a_2 .

To find the constants a_1 and a_2 , note that

$$a_0 = 2 = a_1 + a_2 \text{ and } a_1 = 7 = a_1 2 + a_2 (-1).$$

Solving these equations, we find that $a_1 = 3$ and $a_2 = -1$.

Hence, the solution is the sequence $\{a_n\}$ with $a_n = 3 \cdot 2^n - (-1)^n$.

Using Theorem 1

$$\begin{aligned}a_n &= c_1 a_{n-1} + c_2 a_{n-2} \\ a_n &= a_{n-1} + 2a_{n-2} \\ r^2 - c_1 r - c_2 &= 0 \\ c_1 &= 1, c_2 = 2 \\ r^2 - r - 2 &= 0\end{aligned}$$

- Solve the recurrence

$a_n = a_{n-1} + 2a_{n-2}$ given the initial conditions $a_0 = 2, a_1 = 7$.

- **Solution:** Use theorem 1:

- We have $c_1 = 1, c_2 = 2$
- The characteristic equation is:

$$r^2 - r - 2 = 0$$

- Solve it:

$$r = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-2)}}{2(1)} = \frac{1 \pm \sqrt{9}}{2} = \frac{1 \pm 3}{2} = 2 \text{ or } -1.$$

- so, $r = 2$ or $r = -1$.
- So, $a_n = \alpha_1 2^n + \alpha_2 (-1)^n$.

(Using the quadratic formula here.)

$$\begin{aligned}ax^2 + bx + c &= 0 \Leftrightarrow \\ x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}\end{aligned}$$

Example Cont...

- To find α_1 and α_2 , just solve the equations for the initial conditions a_0 and a_1 :

$$a_0 = 2 = \alpha_1 2^0 + \alpha_2 (-1)^0$$

$$a_1 = 7 = \alpha_1 2^1 + \alpha_2 (-1)^1$$

Simplifying, we have the pair of equations:

$$2 = \alpha_1 + \alpha_2$$

$$7 = 2\alpha_1 - \alpha_2$$

which we can solve easily by substitution:

$$\alpha_2 = 2 - \alpha_1; \quad 7 = 2\alpha_1 - (2 - \alpha_1) = 3\alpha_1 - 2;$$

$$9 = 3\alpha_1; \quad \alpha_1 = 3; \quad \alpha_2 = -1.$$

- Using α_1 and α_2 in $a_n = \alpha_1 2^n + \alpha_2 (-1)^n$, then our final answer is:

$$a_n = 3 \cdot 2^n - (-1)^n$$

Check: $\{a_{n \geq 0}\} = 2, 7, 11, 25, 47, 97 \dots$

The Solution when there is a Repeated Root

Theorem 2: Let c_1 and c_2 be real numbers with $c_2 \neq 0$. Suppose that $r^2 - c_1r - c_2 = 0$ has **one repeated root** r_0 . Then the sequence $\{a_n\}$ is a solution to the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} \text{ if and only if}$$

$$a_n = \alpha_1 r_0^n + \alpha_2 n r_0^n$$

$$a_n = \alpha_1 r_0^n + \alpha_2 n r_0^n, \text{ for all } n \geq 0,$$

for $n = 0, 1, 2, \dots$, where α_1 and α_2 are constants.

Using Theorem 2

Example: What is the solution to the recurrence relation $a_n = 6a_{n-1} - 9a_{n-2}$ with $a_0 = 1$ and $a_1 = 6$?

Solution: The characteristic equation is $r^2 - 6r + 9 = 0$. The only root is $r = 3$. Therefore, $\{a_n\}$ is a solution to the recurrence relation if and only if

$$a_n = a_1 3^n + a_2 n(3)^n$$

where a_1 and a_2 are constants.

To find the constants a_1 and a_2 , note that

$$a_0 = 1 = a_1 \quad \text{and} \quad a_1 = 6 = a_1 \cdot 3 + a_2 \cdot 3.$$

Solving, we find that $a_1 = 1$ and $a_2 = 1$.

Hence,

$$a_n = 3^n + n3^n.$$

More Examples

- $a_n = (1.02)a_{n-1}$
linear
constant coefficients
homogeneous
degree 1
- $a_n = (1.02)a_{n-1} + 2^{n-1}$
linear
constant coefficients
nonhomogeneous
degree 1
- $a_n = a_{n-1} + a_{n-2} + a_{n-3} + 2^{n-3}$
linear
constant coefficients
nonhomogeneous
degree 3
- $a_n = ca_{n/m} + b$
does not have the right form
- $a_n = na_{n-1} + n^2a_{n-2} + a_{n-1}a_{n-2}$
nonlinear
coefficients are not constants
homogeneous
degree 2

Solution Procedure: 2-LiHoReCoCo

2-LiHoReCoCo: • Linear • constant coefficients • homogeneous • degree 2	$a_n = c_1 a_{n-1} + c_2 a_{n-2}$
	Characteristic equation (C.E.): $r^2 - c_1 r - c_2 = 0 \quad (1)$
	The solutions to the RR are given by substituting characteristic roots (CR) from 1 in the following: $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n \text{ for } n \geq 0 \quad (2)$
	Use the initial conditions (e.g. a_0 and a_1) to find α_1 and α_2 in the RR (3)
	Final Result=Use step (2) which is the solution of RR with CR and from (3) solved values for α_1 and α_2 .
	Check your final result to make sure it is correct

Linear
 Homogeneous
 Recurrence of
 degree 2 with
 Constant
 Coefficient

Solving Linear Homogeneous Recurrence Relations of Arbitrary Degree

This theorem can be used to solve **linear homogeneous** recurrence relations with **constant coefficients** of **any degree** when the characteristic equation has **distinct roots**.

Theorem 3: Let $c_1, c_2, \dots, c_k$ be real numbers. Suppose that the characteristic equation

$$r^k - c_1 r^{k-1} - \dots - c_k = 0$$

has k distinct roots $r_1, r_2, \dots, r_k$. Then a sequence $\{a_n\}$ is a solution of the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$$

if and only if

$$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n + \dots + \alpha_k r_k^n$$

for $n = 0, 1, 2, \dots$, where $\alpha_1, \alpha_2, \dots, \alpha_k$ are constants.

Solving Linear Homogeneous Recurrence Relations of Arbitrary Degree

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$$r^k - c_1 r^{k-1} - \dots - c_k = 0$$

has **k distinct** roots $r_1, r_2, \dots, r_k$. Then a sequence $\{a_n\}$ is a solution of the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$$

if and only if

$$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n + \dots + \alpha_k r_k^n$$

for $n = 0, 1, 2, \dots$, where $\alpha_1, \alpha_2, \dots, \alpha_k$ are **constants**.

Using Theorem 3

Find the solution to the RR $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$
with the initial conditions $a_0 = 2, a_1 = 5, a_2 = 15$

The General Case with Repeated Roots Allowed

Theorem 4: Let $c_1, c_2, \dots, c_k$ be real numbers. Suppose that the characteristic equation

$$r^k - c_1 r^{k-1} - \dots - c_k = 0$$

has t distinct roots $r_1, r_2, \dots, r_t$ with multiplicities $m_1, m_2, \dots, m_t$, respectively so that $m_i \geq 1$ for $i = 1, 2, \dots, t$ and $m_1 + m_2 + \dots + m_t = k$. Then a sequence $\{a_n\}$ is a solution of the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$$

if and only if

$$\begin{aligned} a_n = & (\alpha_{1,0} + \alpha_{1,1}n + \dots + \alpha_{1,m_1-1}n^{m_1-1})r_1^n \\ & + (\alpha_{2,0} + \alpha_{2,1}n + \dots + \alpha_{2,m_2-1}n^{m_2-1})r_2^n \\ & + \dots + (\alpha_{t,0} + \alpha_{t,1}n + \dots + \alpha_{t,m_t-1}n^{m_t-1})r_t^n \end{aligned}$$

for $n = 0, 1, 2, \dots$, where $\alpha_{i,j}$ are constants for $1 \leq i \leq t$ and $0 \leq j \leq m_i - 1$.

Using Theorem 4

Example 7: Suppose that the roots of the characteristic equation of a **linear homogeneous recurrence relation** are 2, 2, 2, 5, 5, and 9 (that is, there are three roots, the root 2 with multiplicity three, the root 5 with multiplicity two, and the root 9 with multiplicity one).

What is the form of the general solution?

Solution:

By Theorem 4, the general form of the solution is

Using Theorem 4

EXAMPLE 8 Find the solution to the recurrence relation

$$a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}$$

with initial conditions $a_0 = 1$, $a_1 = -2$, and $a_2 = -1$.

Homogeneous & Nonhomogenes

Homogeneous:

If we put all the a_i 's on one side of the equation and everything else on the right side, then the right side is 0.

Otherwise **inhomogeneous** or **nonhomogeneous**.

It is also homogenous because in the following formula **no** terms occur that are **not** multiples of a_j 's.

The **coefficients** of the terms of the sequence are all **constants**, rather than **functions that depend on n** .

Linear Nonhomogeneous Recurrence Relations with Constant Coefficients (LiNoReCoCos)

Definition: A *linear nonhomogeneous recurrence relation with constant coefficients* is a recurrence relation of the form:

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} + F(n),$$

where $c_1, c_2, \dots, c_k$ are real numbers, and $F(n)$ is a function not identically zero depending only on n .

The recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k},$$

is called the *associated homogeneous recurrence relation*.

LiNoReCoCos – cont.

- Linear *nonhomogeneous* RRs with constant coefficients may (unlike LiHoReCoCos) contain some terms $F(n)$ that depend *only* on n (and *not* on any a_i 's). General form:

$$a_n = c_1 a_{n-1} + \dots + c_k a_{n-k} + F(n)$$


The *associated homogeneous recurrence relation* (associated LiHoReCoCo).

LiNoReCoCos – cont.

The following are linear **nonhomogeneous** recurrence relations with constant coefficients:

$$a_n = a_{n-1} + 2^n,$$

$$a_n = a_{n-1} + a_{n-2} + n^2 + n + 1,$$

$$a_n = 3a_{n-1} + n3^n,$$

$$a_n = a_{n-1} + a_{n-2} + a_{n-3} + n!$$

where the following are the associated linear **homogeneous** recurrence relations, respectively:

$$a_n = a_{n-1},$$

$$a_n = a_{n-1} + a_{n-2},$$

$$a_n = 3a_{n-1},$$

$$a_n = a_{n-1} + a_{n-2} + a_{n-3}$$

Solving Linear Nonhomogeneous Recurrence Relations with Constant Coefficients

Theorem 5: If $\{a_n^{(p)}\}$ is a **particular solution** of the nonhomogeneous linear recurrence relation with constant coefficients

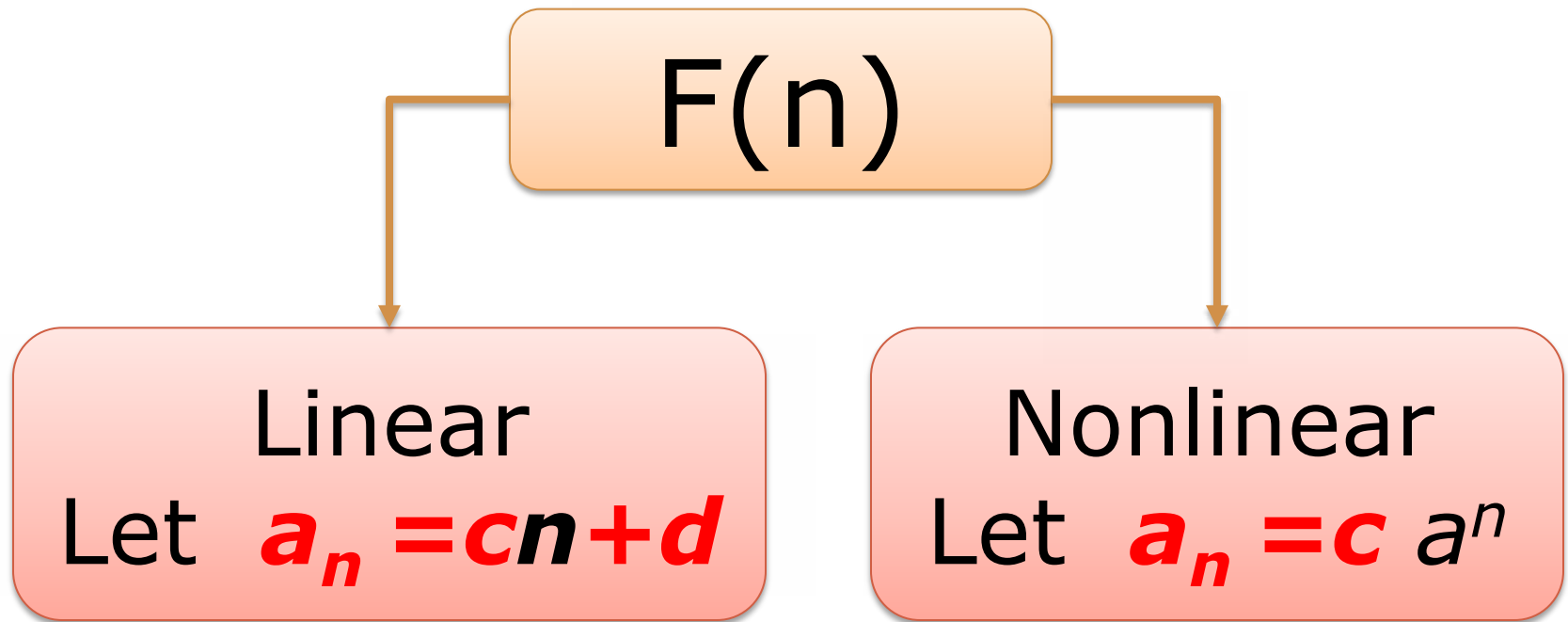
$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} + F(n),$$

then every solution is of the **form** $\{a_n^{(p)} + a_n^{(h)}\}$, where $\{a_n^{(h)}\}$ is a **solution of the associated homogeneous** recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}.$$

$$\textbf{Solution} = a_n = \{a_n^{(p)} + a_n^{(h)}\}$$

F(n) Types



When $F(n)$ is linear

Linear
Let $a_n = cn + d$

Find all solutions of the recurrence relation $a_n = 3a_{n-1} + 2n$. What is the solution with $a_1 = 3$?

When $F(n)$ is nonlinear

Nonlinear
Let $a_n = c a^n$

Find all solutions of the recurrence relation

$$a_n = 5a_{n-1} - 6a_{n-2} + 7^n.$$